

Workshop 2

Labeling Trades and Forming Teams

Give every agent decision an outcome you can check, then pick a lane and a team.

Travis Sim

NUS · Singapore

Thu 5 Nov 2026 · COM3/COM4

Where we are

Today's question: how do we know if a decision was any good?

DONE

Build night 2

Thu 29 Oct

Data bars and change alerts

TODAY

Workshop 2

Thu 5 Nov

Labeling trades and forming teams

NEXT

Build night 3

Thu 12 Nov

Position sizing and the trade ledger

Tonight

18:00 to 20:00. The first half is about labels, the second half is about your team.

18:10 — 40 min
Teaching
labels, leaks, the gate, lanes

18:50 — 10 min
Live demos
three demos on real prices

19:00 — 25 min
Team matching
pitch, choose, find, agree

19:25 — 10 min
Mini-lab
label ten decisions by hand

19:35 — 20 min
Open build
your own project

19:55 — 5 min
Push
one line on the next step

A decision needs a checkable outcome

Until you can say "that one worked", you cannot test an agent.

THE AGENT SAYS

Buy SOL. It rose 6 bps in the last minute.



WHAT WE NEED

A label: right, wrong, or nothing happened.

right

wrong

nothing happened

A label turns a decision into a row you can count, chart and test.

Illustrative

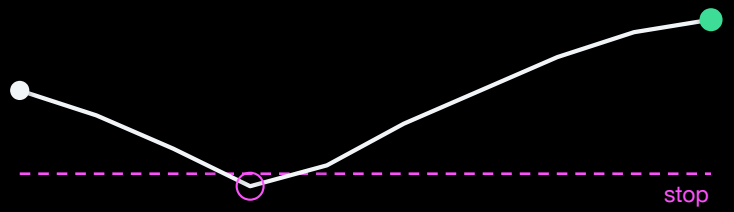
Fixed-time labels are crude

"Look 2 minutes later" ignores what happened on the way.

PATH A

fixed label: +

Falls 6 bps first, then recovers



PATH B

fixed label: +

Climbs slowly and steadily

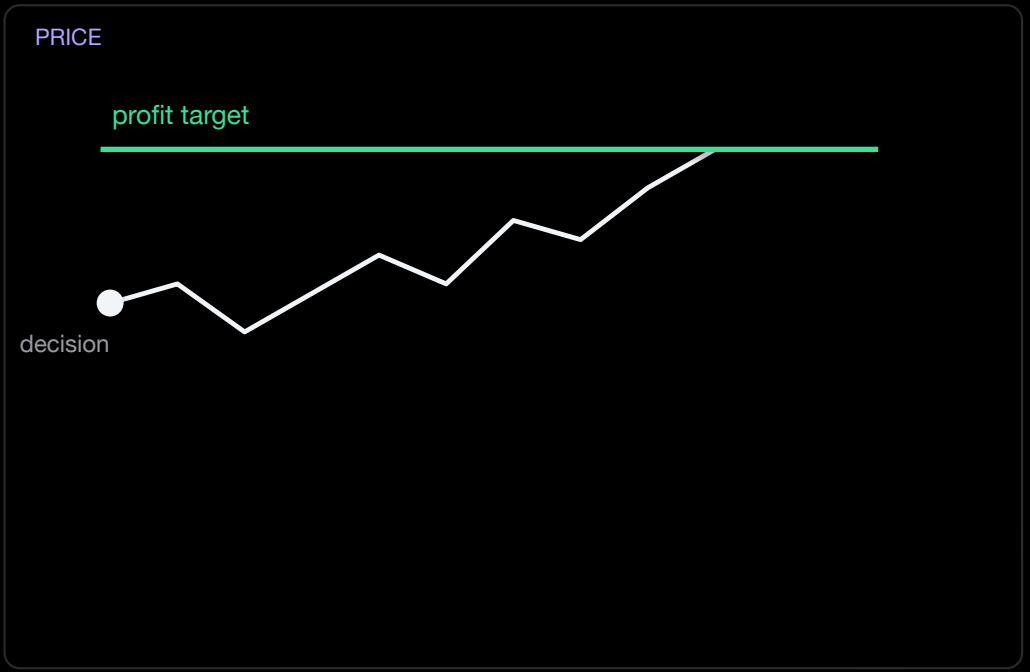


A fixed-time label calls both "+". A trader with a stop was already out of A.

Illustrative paths

The triple barrier

Put walls around the decision.



STEP 1 OF 3

A profit target above the entry price. Touch it first and the decision worked.

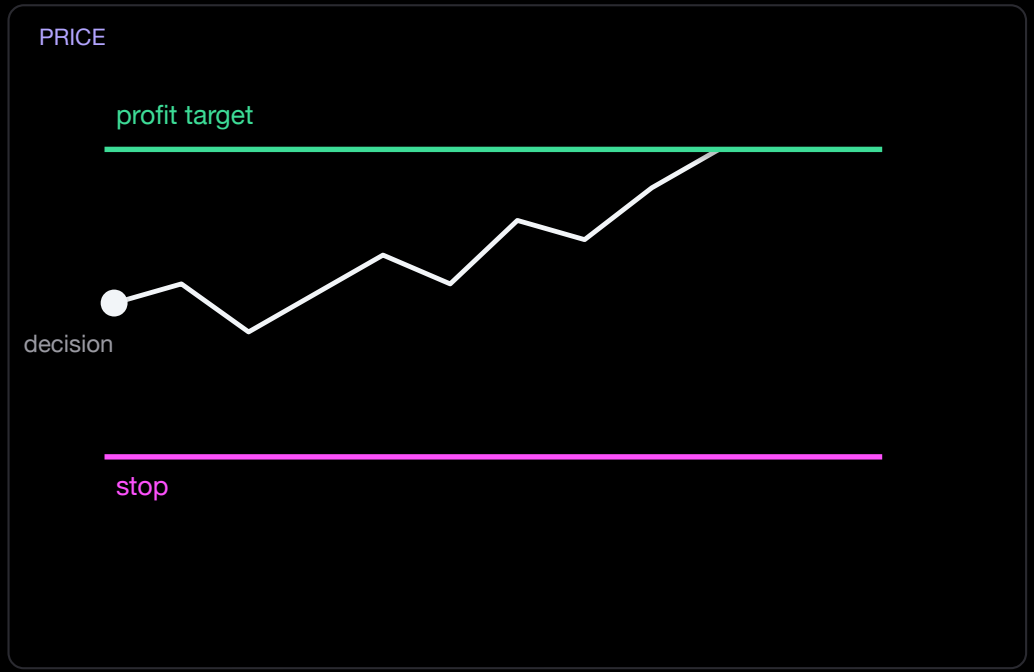
LABEL RULE

```
upper barrier -> label +1
```

Illustrative price path

The triple barrier

Put walls around the decision.



STEP 2 OF 3

A stop below the entry price. Touch it first and the decision failed.

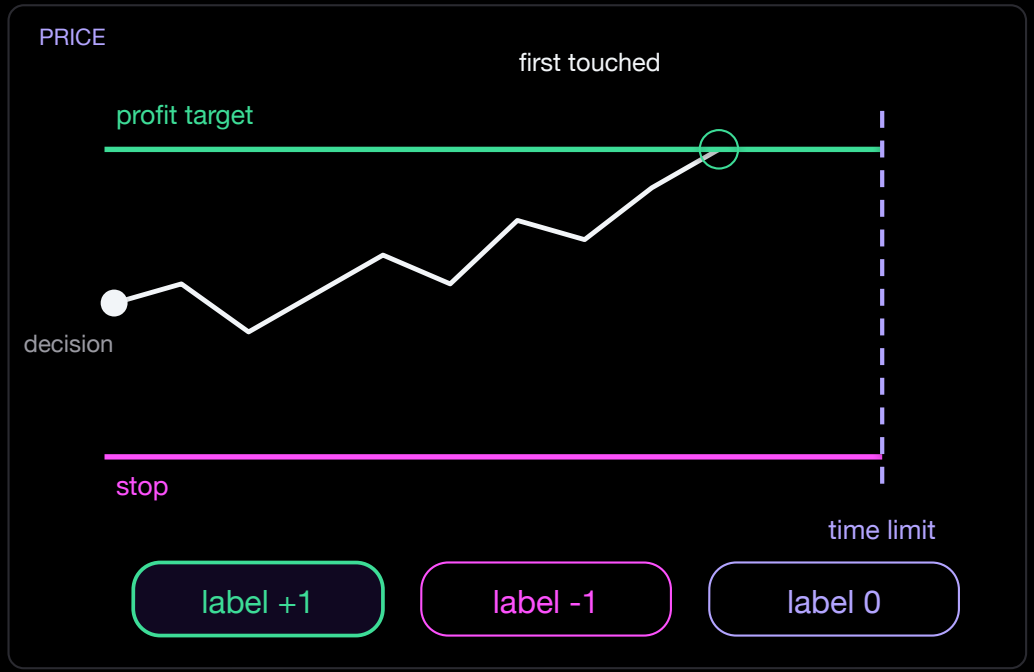
LABEL RULE

```
lower barrier -> label -1
```

Illustrative price path

The triple barrier

The first wall the price touches is the label.



STEP 3 OF 3

A time limit on the right. If neither wall is touched, nothing happened.

LABEL RULE

```
time limit -> label 0
```

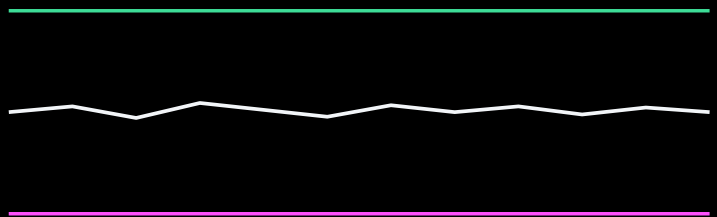
Illustrative price path

Barrier width follows the noise

A fixed width is too wide on a quiet day and too narrow on a busy one.

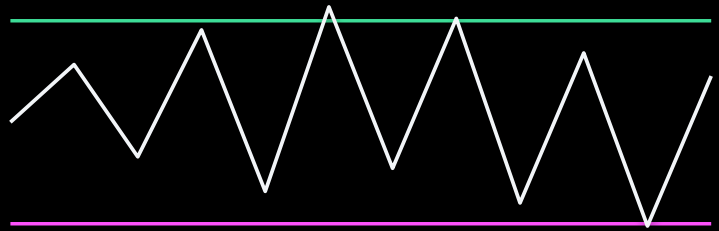
QUIET

Same width: the walls are never touched



BUSY

Same width: touched in seconds, by noise



THE RULE WE USE

```
width = max(floor, 0.8 x noise x sqrt(24 steps))
```

One real decision: noise 0.92 bps per step, 24 steps, so 0.8 x 0.92 x 4.9 = 3.6 bps.

The floor (2 bps) stops a flat stretch from making the walls touch the price.

Illustrative; the example numbers are from out/labels.json

A win has to beat the cost

A label of +1 is gross. Fees come off afterwards.



ROUND TRIP IN POOL FEES

0.04% each way

0 of 73

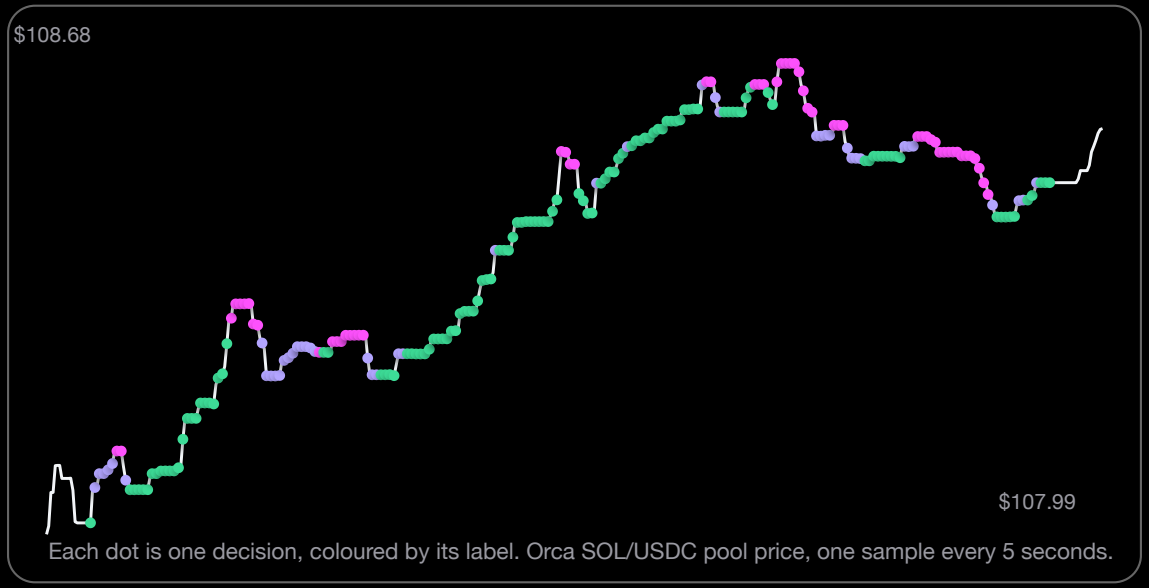
winning labels had a barrier at least 8 bps wide.

A "win" narrower than the cost still loses money. Set the width floor to your costs.

Orca SOL/USDC pool fee 0.04% (recorded/SOURCES.md); widths from out/labels.json and out/agent.json

Real labels on real prices

SOL/USDC, 40 minutes, a decision every 10 seconds.



Each dot is one decision, coloured by its label. Orca SOL/USDC pool price, one sample every 5 seconds.

PROFIT TARGET FIRST (+1)

118 — 54% of 219

STOP FIRST (-1)

56 — 26% of 219

TIME RAN OUT (0)

45 — 21% of 219

A fixed-time label disagrees on 57 of 219 (26%).

Reads use public mainnet data; every write is devnet only; no real funds. Simulated, on recorded data; not advice.

Source: out/labels.json, recorded Orca SOL/USDC pool price, 21 Oct 2026 17:14 to 17:54 UTC, 481 samples (from build night 1). Gross of fees.

Labels depend on the price path

Same agent, same settings, three real price paths.

PRICE PATH	UP	DOWN	TIME	WIDTH	TRADES	HIT RATE
Pyth oracle	86	45	88	7.4 bps	67	14.9%
Jupiter quote	94	58	67	7.2 bps	75	32%
Orca pool price	118	56	45	3.4 bps	143	51%

Pyth updates about every 10 seconds and Jupiter quotes include routing noise, so their paths are choppier than the pool price. Pick the path your agent really sees.

out/sources.json (gross of fees; simulated, on recorded data; not advice)

The labeler in code

demo-repo/code/src/labels.ts

```
  const last = i + p.horizon;
  if (last > prices.length - 1) return null;
  const price = prices[i]!;
  const widthBps = barrierWidthBps(v, p);
  const up = price * (1 + widthBps / 10_000);
  const dn = price * (1 - widthBps / 10_000);
  let exitI = last;
  let touch: Label["touch"] = "time";
  for (let k = i + 1; k <= last; k++) {
    const x = prices[k]!;
    if (x >= up) { exitI = k; touch = "up"; break; }
    if (x <= dn) { exitI = k; touch = "down"; break; }
  }
```

1 **Walk forward**

Step by step, up to the time limit.

2 **First touch wins**

The loop stops at the first barrier.

3 **Not known yet**

No label if the time limit runs past the data.

Source: demo-repo/code/src/labels.ts, function labelAt (comments and blank lines removed)

Labels overlap

Decide often, look far ahead: the labels share the same future.



A decision every 20 seconds, each looking 100 seconds ahead (illustrative)

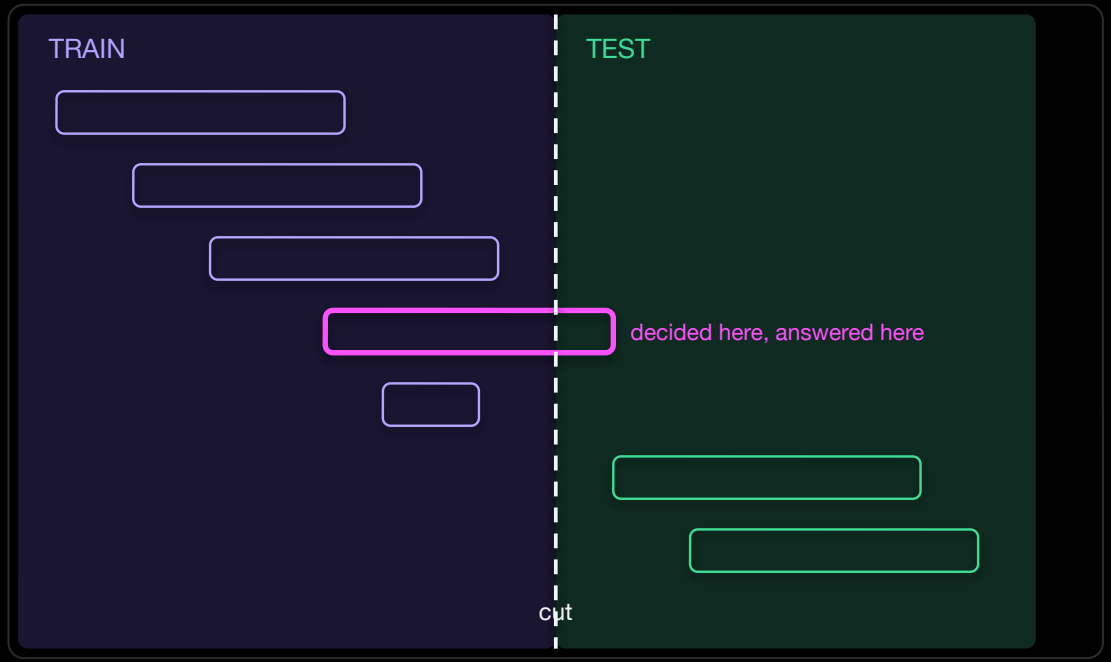
Three labels, one price path. The shaded part is future that two or more labels use.

Each label feels like a new fact. Most of it is the same fact.

Illustrative

Overlap leaks into the test

A label decided before the cut can use prices after the cut.



The crossing label was decided in the train period, but its answer comes from test prices.

THE FIX

Purge the labels that cross the cut before you fit anything.

Illustrative

How much is really new?

Real labels from the 40-minute recording.

ALIVE AT ONCE

6.9

on average, at most 13

AVERAGE UNIQUENESS

16%

of a label is not shared

INDEPENDENT LABELS

34

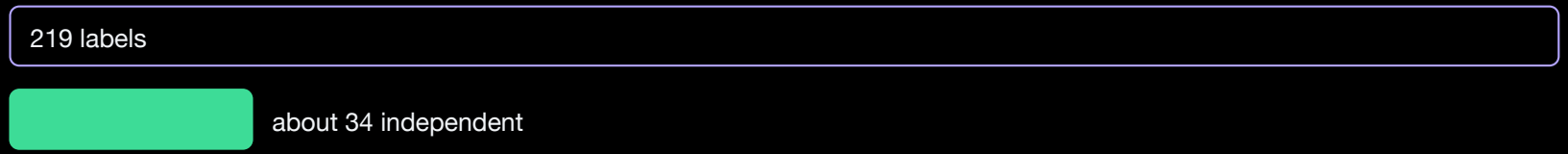
worth of new facts in 219 labels

CROSS THE CUT

9

training labels at step 282 of 481

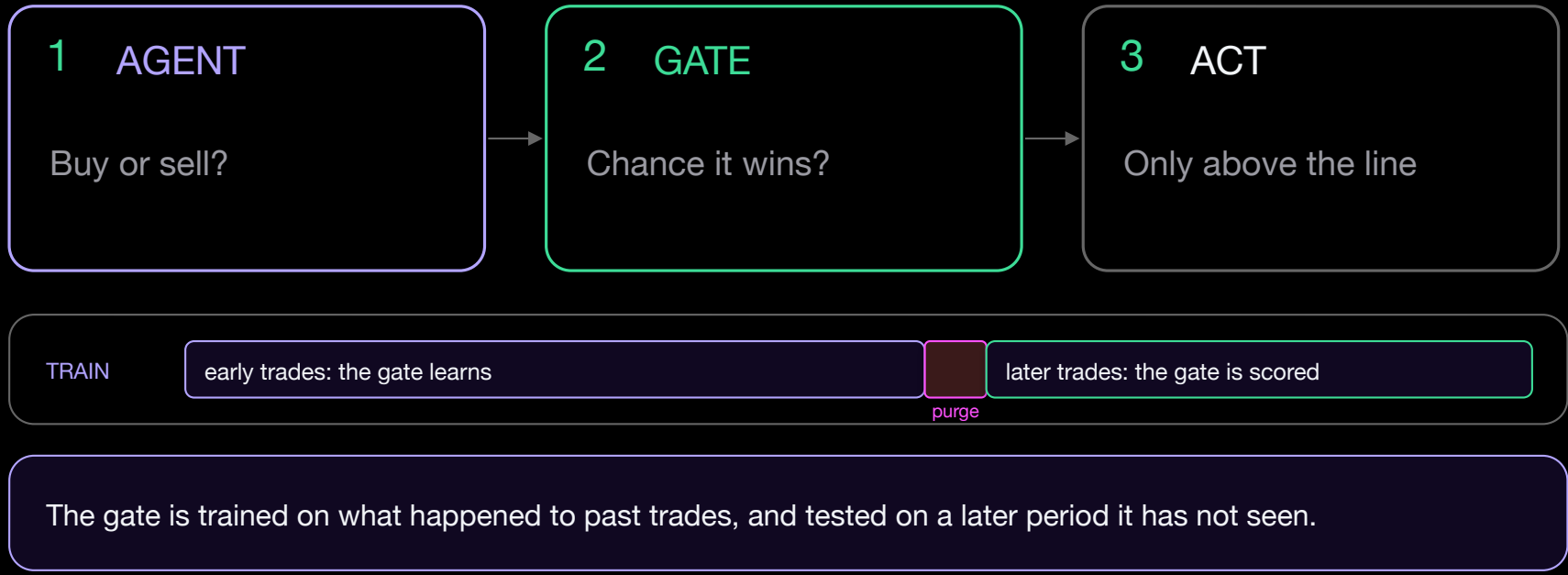
LABELS VERSUS INDEPENDENT LABELS



out/overlap.json (time limit 24 steps, a decision every 2 steps, cut at 60% of the labels)

A second opinion

The agent picks the side. A second check decides whether to act.



Illustrative flow

The pretend agent writes reasons

No AI credits, no network: a rule that explains itself.

A WRITTEN REASON

"Buy: SOL fell 9.7 bps in the last 60 s, 2.2x the recent noise, so follow the move. The two venues agree within 4.0 bps."

BUY

one claim in this reason was given a simulated error

The wrong-claim rate is 20%, fixed seed, flagged in out/agent.json.

THE CHECKER READS THE FACTS

Direction	says fell	data: rose	WRONG
Size	says 9.7 bps	data: 9.7 bps	TRUE
Noise ratio	says 2.2x	data: 2.2x	TRUE
Venue gap	within 4.0 bps	data: 4.0 bps	TRUE

It checks direction, size and the venue gap. It does not judge the idea.

out/agent.json

Grade the reasons, then the result

Two separate questions, and the answers do not have to agree.

QUESTION 1: WAS THE REASON TRUE?

43 of 43

simulated wrong claims caught

0 false alarms on 176 true reasons. Easy: we made the errors ourselves.

QUESTION 2: DID THE TRADE WIN?

Reason was true

51.3% hit rate, 113 trades

Reason had a wrong claim

50% hit rate, 30 trades

A true reason did not make a better trade: 51.3% versus 50%. Grade the facts first, then the label.

out/agent.json (gross of fees; simulated, on recorded data; not advice)

The gate on the test period

Trained on early trades, scored on later ones, with a +/- range.

HIT RATE ON 55 TEST TRADES, WITH A +/- RANGE

No gate: 55 trades, 45.5% (+/- 13.2)



With the gate: 47 trades, 48.9% (+/- 14.3)



BLOCKED BY THE GATE

8 trades

2 of them would have won

Trained on 82 trades; 6 more were purged for crossing the cut.

With this few trades the +/- ranges overlap. This is a method to practise, not a result to trust.

out/meta.json (simulated, on recorded data; not advice)

An on-chain record

Write each decision to devnet with the Memo program: a record outside your laptop.

MEMO LINE

{"app":"labeler","i":20,"t":1792602939,"side":"sell","label":1,"h":"b77d9610847aaf99"}

HOW IT IS BUILT

step, side, label and a hash of the full record. One signer, one Memo instruction, hand-built, Node ed25519.

DEVNET DRY RUN (simulateTransaction)

```
Program Memo...GmfcHr invoke [1]
Program log: Memo (len 86): "{...}"
Program Memo...GmfcHr success
```

3 of 3 accepted, err null, 44,922 to 45,286 compute units of 200,000

Status: dry run on devnet works. Sending needs a funded devnet key: ask a lead.

recorded/devnet-simulation.json (simulateTransaction on devnet, nothing sent)

Three lanes

Solo or a team of 3 to 4. Pick one lane for the final project.

AGENT

A trading agent or strategy with honest validation.

EXCHANGE

A venue component: order book, matching, deposit or custody flow.

TOOLING

Anything that helps the trading process: data, monitoring, risk, explainability, dashboards.

Judged on utility, tests, devnet evidence and a clear explanation. Never on simulated returns.

Starters you can run today

Each one has a first test you can measure.

AGENT

Label my own strategy	BEG
Build the second-opinion gate	STD
Explanation checker for an LLM agent	STD
Agent with a risk hook and a label log	ADV

EXCHANGE

Deposit watcher	BEG
Mini order book	STD
Orca Whirlpool devnet lab	ADV
Read a real on-chain order book (Phoenix)	ADV

TOOLING

Data collector for another market	BEG
Change and staleness monitor	STD
Label and gate dashboard	STD
Leak checker for backtests	ADV

Suggestions, not a menu. Bring your own idea if you have one.

demo-repo/code/data/lanes.json (from research/experiments, checked 20-21 Oct 2026)

The one-page brief

Nine lines. If you cannot fill one, that is the first task.

1 **User**
who has this problem

2 **Observed failure**
and your evidence

3 **Smallest useful component**
what you will build first

4 **Success test**
measurable

5 **Devnet component**
the on-chain evidence

6 **Limitations**
what it will not do

7 **Contributions**
what each person brings

8 **Learning goals**
what each person wants to learn

9 **First tasks**
three, with names

Plus how you will talk to each other and when you are free.

Team matching in 25 minutes

About 10 students: aim for two or three teams. Solo is fine.

0-5 min

Pitch

one problem each, with evidence you saw

5-10 min

Choose

first interest, second choice

10-20 min

Find

share what you bring and want to learn

20-25 min

Agree

provisional team, shared problem, first tasks

Interest card: handle | first and second problem | skill I bring | skill I want to learn | availability

Teams can change by agreement. We check fit again on Thu 12 Nov and Thu 21 Jan.

Triple barrier on real prices

PREDICT FIRST

If you double the width, do more labels become +1 or 0?

```
$ npm run demo -- --recorded --mult 1.6
```

YOU SHOULD SEE

- Fewer touches, more "time ran out"
- The average width roughly doubles

Fallback: screenshot below; the lab page works from disk



Learning Lab demo, runs from disk, recorded data.

`Try it: nus-solana-demos.vercel.app/0004/01-triple-barrier.html`

Live mode: switch it on to watch real Solana data.

Source: demo-repo/lab/screenshots (captured from the Learning Lab demos on recorded data)

Overlap and leakage

PREDICT FIRST

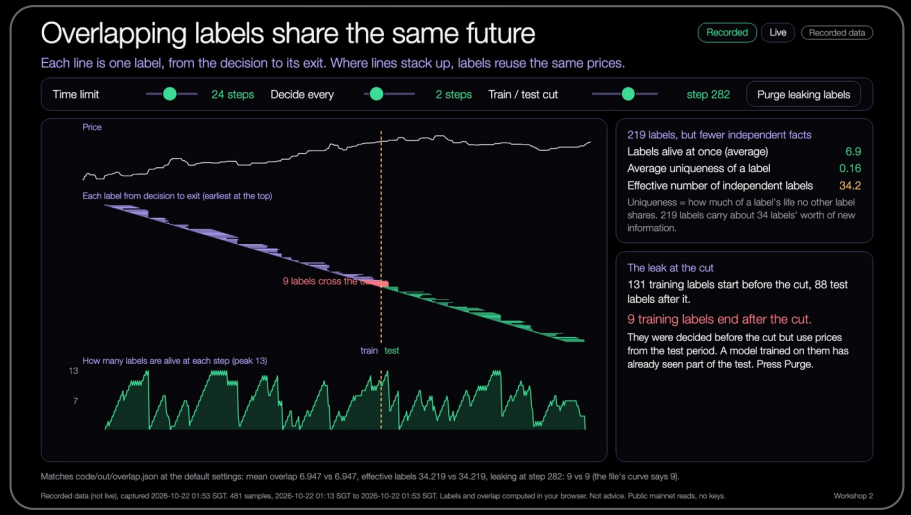
Where should the cut go so that nothing leaks?

```
$ open demo-repo/lab/02-overlap-leakage.html
```

YOU SHOULD SEE

- Around 7 labels alive at once
- Purge removes the crossing labels

Fallback: screenshot below; the lab page works from disk



Learning Lab demo, runs from disk, recorded data.

`Try it: nus-solana-demos.vercel.app/0004/02-overlap-leakage.html`

Live mode: switch it on to watch real Solana data.

Source: demo-repo/lab/screenshots (captured from the Learning Lab demos on recorded data)

Demo 3

Agent report card

PREDICT FIRST

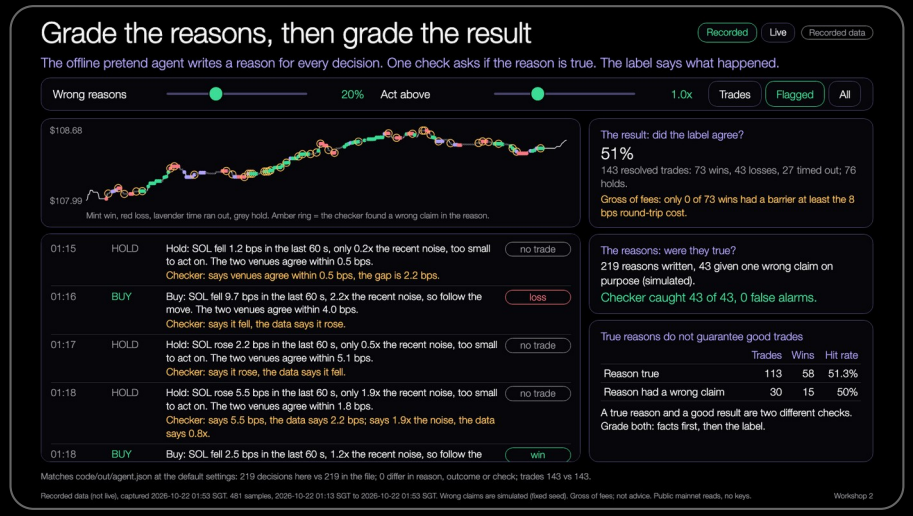
Do the trades with true reasons win more often?

```
$ npm run demo -- --recorded --error-rate 40
```

YOU SHOULD SEE

- A flagged reason shows what is wrong
- True reasons do not guarantee wins

Fallback: screenshot below; the lab page works from disk



Learning Lab demo, runs from disk, recorded data.

Try it: nus-solana-demos.vercel.app/0004/03-agent-report-card.html

Live mode: switch it on to watch real Solana data.

Source: demo-repo/lab/screenshots (captured from the Learning Lab demos on recorded data)

The same agent, with a gate

PREDICT FIRST

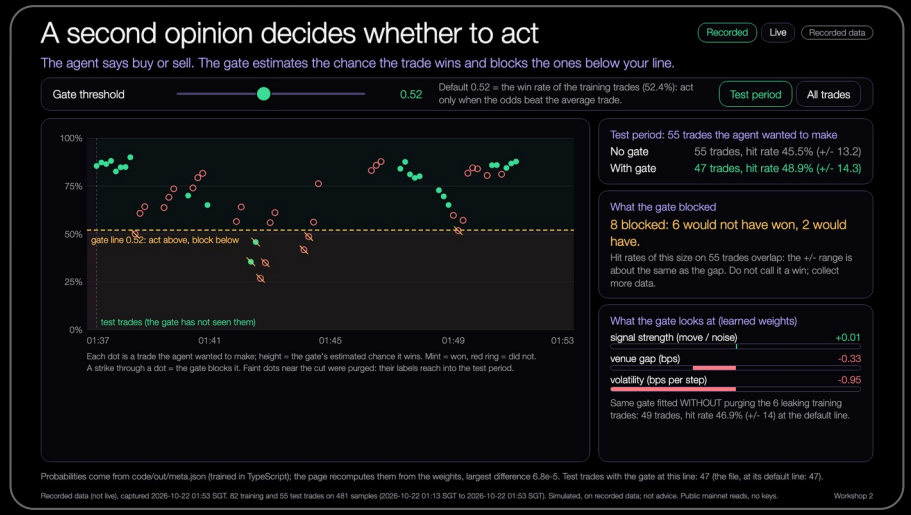
Which blocked trades would have won?

```
$ npm run demo -- --recorded --train-frac 0.5
```

YOU SHOULD SEE

- Blocked trades, some of them winners
- A +/- range as wide as the gap

Fallback: screenshot below; the lab page works from disk



Learning Lab demo, runs from disk, recorded data.

`Try it: nus-solana-demos.vercel.app/0004/04-meta-gate.html`

Live mode: switch it on to watch real Solana data.

Source: demo-repo/lab/screenshots (captured from the Learning Lab demos on recorded data)

Lane and team picker

PREDICT FIRST

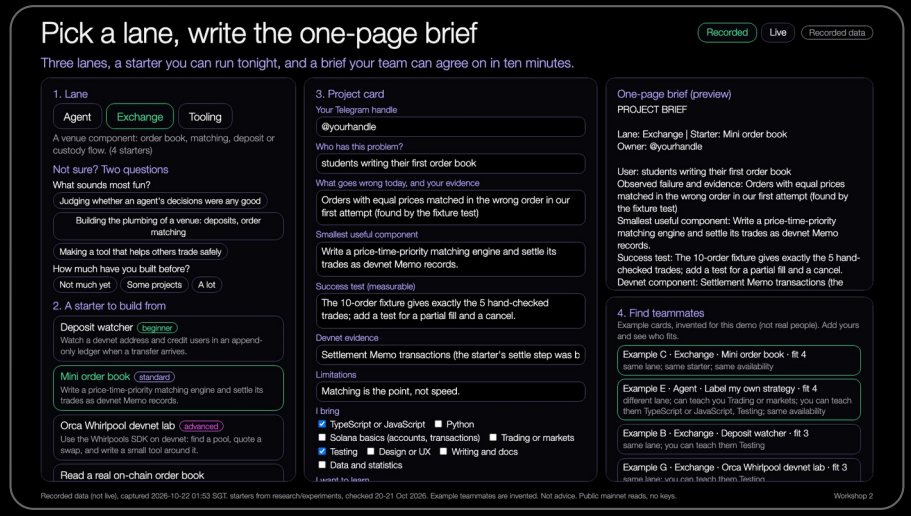
Who fits your card best?

```
$ open demo-repo/lab/05-lane-team-picker.html
```

YOU SHOULD SEE

- A starter fills in the brief
- Example teammates ranked by fit

Fallback: screenshot below; the lab page works from disk



Learning Lab demo, runs from disk, recorded data.

Try it: nus-solana-demos.vercel.app/0004/05-lane-team-picker.html

Live mode: switch it on to watch real Solana data.

Source: demo-repo/lab/screenshots (captured from the Learning Lab demos on recorded data)

Try the demos yourself

DEMOS

`nus-solana-demos.vercel.app/0004/`

Five demos. Open them now.

No install. The same files are in the repo and open from disk. Do not all switch on Live at once: we share one connection.



Scan to open the demos

Source: demo-repo/lab (the five Learning Lab demos, deployed as static pages, recorded data)

Mini-lab: label ten decisions by hand

In your new teams. Ten minutes.

1

Open the sheet

`out/minilab.md` has ten real decisions

2

Label them

+1, -1 or 0 from the price lists

3

Check the key

`out/minilab-answers.md`

4

Change one thing

`npm run demo -- --mult 1.5` and relabel decision 1

Done when your team agrees on all ten and can say why decision 1 changed.

Build menu

Pick one and make it yours.

BEGINNER	Label your own strategy's past decisions
BEGINNER	Build volatility-scaled barriers for another token
STANDARD	Build the gate with your own features
STANDARD	Build an explanation checker that flags reasons contradicting the data
ADVANCED	Label your Session 0 strategy and log it to devnet

Three things to keep

1. A label turns a decision into a checkable outcome.
2. Overlapping labels leak: purge before you test.
3. Your team brief is the first thing you build.

Branch out

Take it further after tonight.

BEGINNER

Change the width, the time limit and the floor. Write down what moves.

STANDARD

Label your own decisions with labelAt, or add a feature to the gate and compare with the +/- range.

ADVANCED

Grade the reasons of your own LLM agent, write a purged k-fold split, or log labels to devnet.

Share back

Show the room, and the next room.

BY FRI 6 NOV

1

Post your one-page brief in the group

THU 12 NOV

2

Bring your first task done to Build night 3

THU 21 JAN

3

Teams checked again before Workshop 3

Get the code

REPO

`github.com/travisim/solana-agentic-trading-nus`

`folder: 0004-labeling-trades-forming-teams/demo`

```
1  $ cd demo/code
```

```
2  $ npm install
```

```
3  $ npm run demo -- --recorded
```

Reads use public mainnet data; every write is devnet only; no real funds.

Scan for the repo folder

Let's connect



LinkedIn

linkedin.com/in/travisim



Telegram

t.me/travisim


SOLANA